

Trainings ▾

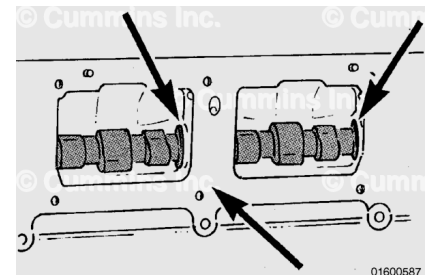
Inspect

Inspect the outside surfaces of the cylinder block and inside the cylinder block around the camshaft bores for discoloration.

Discoloration from high temperature indicates the need for a thorough inspection. Repair the camshaft bore if it is deemed necessary. Reference the machine section below.

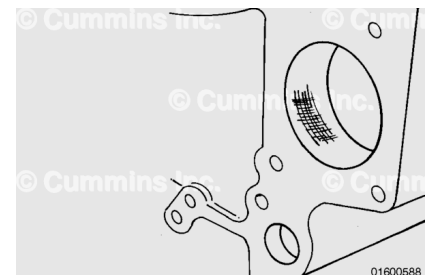
The minimum hardness for the cylinder block is 205HB.

Note : When a hardness test is performed, be sure to use a maximum of 5 dents per test area. An excessive number of dents will result in irreparable block damage.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Remove the camshaft from the cylinder block. Refer to Procedure 001-008 in Section 1.

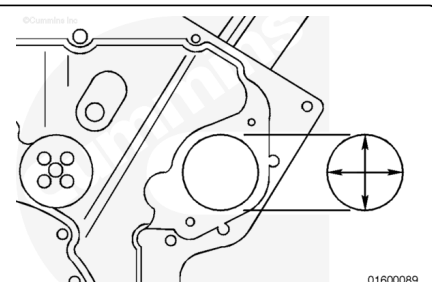
(</qs3/pubsys2/xml/en/procedures/56/56-001-008-tr.html>)

Use solvent to clean the camshaft bushing.

Check the amount of copper exposure on the camshaft bushing. If it extends more than 120 degrees around the inside diameter of the camshaft bushing, the bushing **must** be replaced.

Measure the camshaft bushing inside diameter in four equally spaced locations.

Camshaft bushing specifications:



Camshaft Bushing Inside Diameter

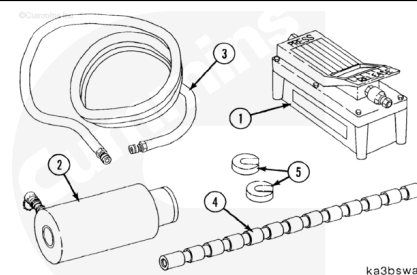
mm		in
105.114	MIN	4.138
105.177	MAX	4.141

If the camshaft bushing is out of specification, the camshaft bushing **must** be replaced.

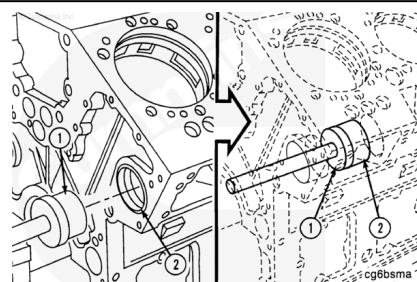
Inspect the camshaft bushing bore while removing the camshaft bushing.

Remove

Use the camshaft bushing tool hydraulic actuator kit, Part Number 3823621, and the camshaft bushing installation/removal kit, Part Number 3163015, to remove the camshaft bushings.



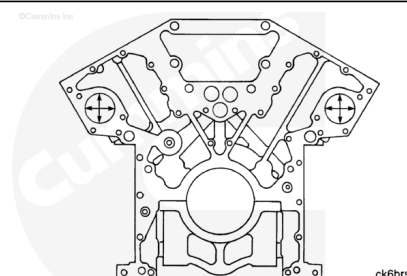
Remove the camshaft bushings.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

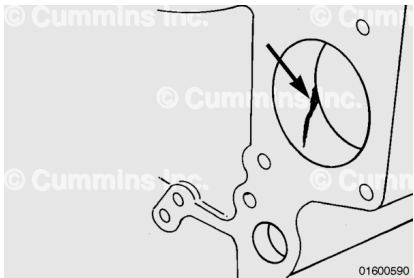


Use solvent to clean the camshaft bores.

Use an emery cloth to remove any burrs and sharp edges from the bore.

Use crack detection kit, part Number 3375432, or equivalent, to check the camshaft bores for cracks.

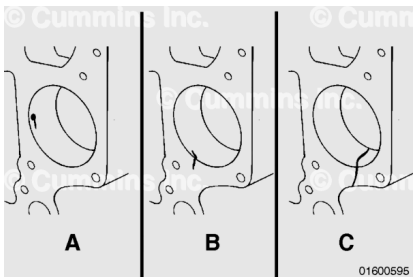
Bores which have excessive and multiple cracks can **not** be repaired. Reference the guidelines below.



Guidelines for unacceptable cracks:

- A. Cracks entering oil hole
- B. Cracks in the radii of the structural webs
- C. Cracks that break over the side of the camshaft bearing bore.

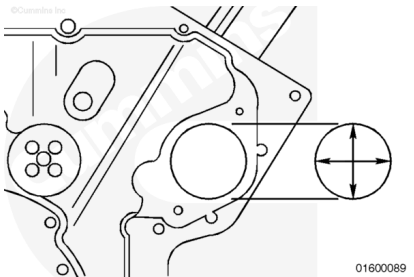
If any camshaft bearing bores contain any of these cracks, the block can **not** be repaired and **must** be replaced.



Measure the camshaft bushing bore inside diameter in four equally spaced locations.

Camshaft bushing bore specifications:

Camshaft Bushing Bore Diameter		
mm		in
111.975	MIN	4.409
112.025	MAX	4.410



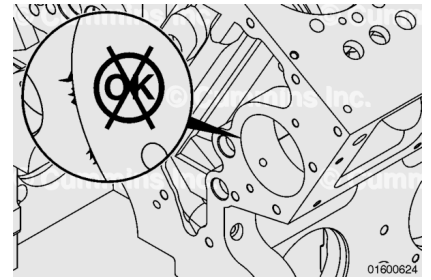
If the camshaft bushing bore is **not** within specifications, the bore **must** be machined and a repair sleeve inserted.

Reference the repair section below for information on salvaging the camshaft bushing bores.

Install

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



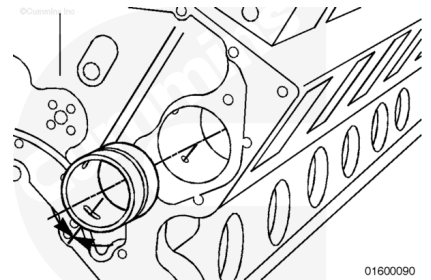
Prior to installing the bushing, ensure there are no burrs around the edge of the camshaft bushing bores.

If found, use an appropriate tool to remove any burrs.

Clean the bushing bores thoroughly and dry with compressed air. Refer to Procedure 001-026 in Section 1.

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Align the oil holes in the bushing with the oil drilling in the block.

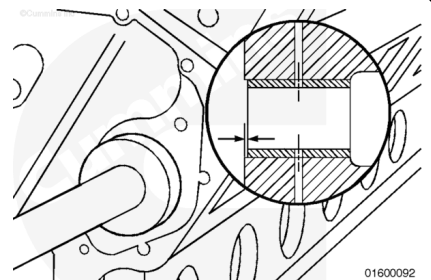


⚠ CAUTION ⚠

Do not damage the bushing during installation. If damage occurs serious engine damage can result.

⚠ CAUTION ⚠

To reduce the possibility of engine damage, make sure that the front camshaft bushings are below the surface of the block.



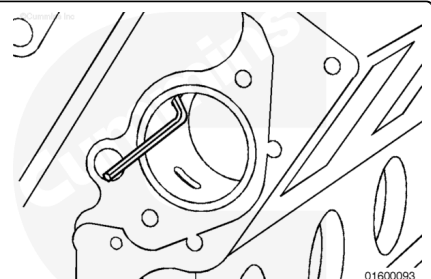
Install the bushing until the oil holes are aligned. Use the camshaft bushing tool hydraulic actuator kit, Part Number 3823621, and the camshaft bushing installation/removal kit, Part Number 3163015, to install the camshaft bushing.

Check the alignment of the bushing oil holes.

Use a 7-mm [1/4-in] Allen wrench. If the Allen wrench will **not** fit into the oil drilling, the camshaft bushing **must** be replaced.

There are three holes in each bushing.

All holes **must** be aligned, except on the front bushing.



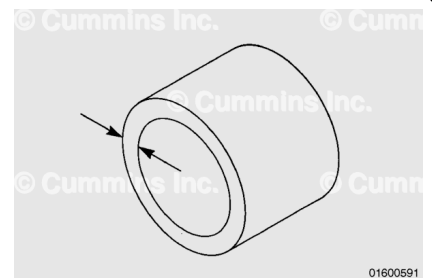
There are no drillings to align with the upper holes on the front camshaft bushing.

Repair

If any of the camshaft bores need repair, the size of the bore **must** be increased so that a repair sleeve can be installed.

There are two repair sleeves available. Reference the Camshaft Bore Repair Sleeves tool in Section 11 of the Service Products Catalog, Bulletin 3377710, for the part numbers for the repair sleeves.

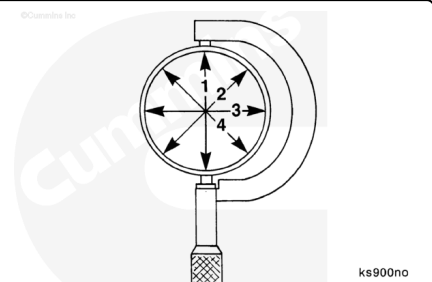
Note : The difference between the repair sleeves is width. Be sure to select the correct repair sleeve for the camshaft bore being repaired.



Measure the outside diameter of the repair bushing in 4 equally spaced places.

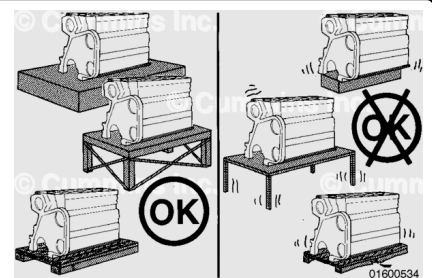
Use this measurement to determine how much material is to be removed during the machining process.

The repair sleeve **must** have an interference fit of 0.063 to 0.139 mm [0.0025 to 0.0055 in]. If the repair sleeve does **not** have an interference fit within this range, the final assembly can **not** be performed.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Put the cylinder block on a vibration-resistant work surface.

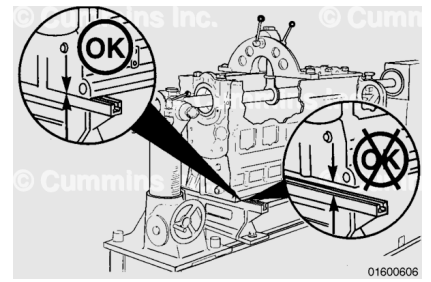
The block **must** sit flat on the surface.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this

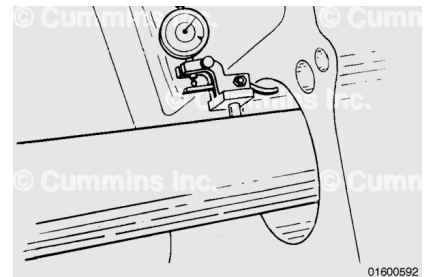
component or assembly.

Position the cylinder block to permit the use of a steady rest during the machining operation.



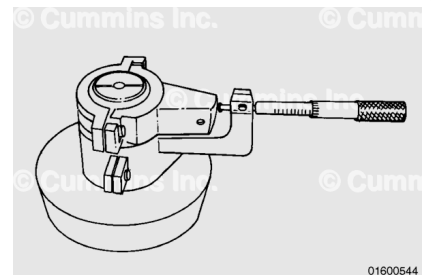
Use a dial indicator to align the center line of the camshaft bore with the center line of the boring bar.

The most accurate alignment is achieved from the camshaft bores that are not damaged and are at the ends of the cylinder block.



Use the average outside diameter measurement value of the camshaft repair sleeve to determine how much material **must** be removed from the camshaft bore.

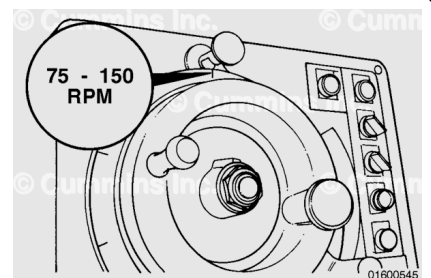
The camshaft bore repair diameter **must** be cut correctly for the repair assembly or repair sleeve to fit correctly.



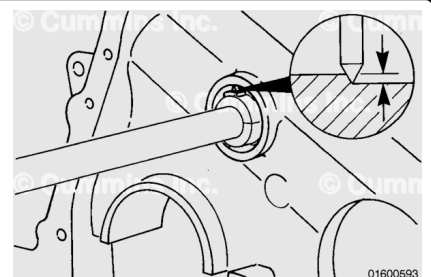
Operate the cutter at 150 to 275 rpm.

If the cutting tool tip wears quickly or breaks, reduce the speed of the cutter.

After cutting the first time, approximately 1.52 mm [0.060 in] of material can be removed from the diameter each additional time the bore is cut.

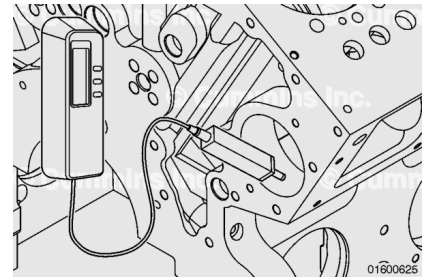


The amount of material that can be removed depends on the harness of the block, the type of boring machine, and the surface finish requirements of the bore.



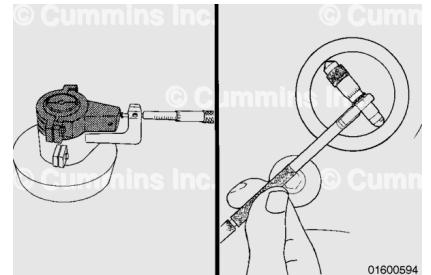
The surface finish of the camshaft bore should be 2.03 μm [80 μin] or less.

Reduce the depth of the cut and reduce the cutter speed, or feed if the surface finish is **not** 2.03 μm [80 μin] or less.



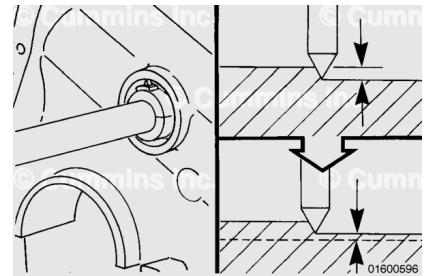
After making the first cut, check the accuracy of the cutting tool setting by comparing the cut diameter of the repair camshaft bore with the set diameter of the cutter.

The difference, if any, is the amount that **must** be taken into consideration when making the final cut. This will allow the bore to be cut to the correct finish.



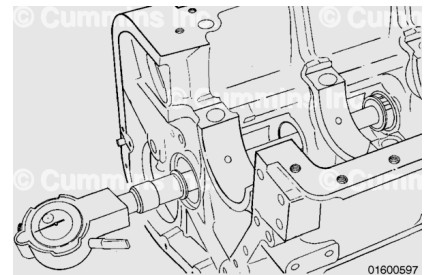
Continue to cut the block until the inside diameter of the camshaft bore is 0.13 mm [0.005 in] smaller than the required bore size.

Reference the measured camshaft bushing repair sleeve outer diameter.

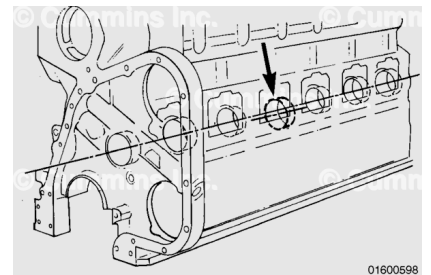


Remove the remaining 0.13 mm [0.005 in] of material from the camshaft bore using reduced depth of cutting operations.

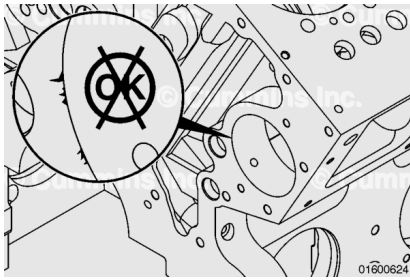
Check the cut diameter to be sure it provides the correct press fit and surface finish.



Complete the cutting operation on each camshaft bore before proceeding to the next bore(s). This lessens the potential for misalignment between adjacent camshaft bores.

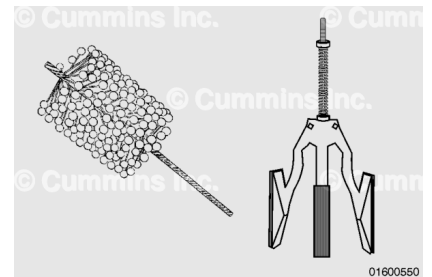


After the cutting operation is complete, use an appropriate tool to remove burrs from around the edges of the camshaft bores.



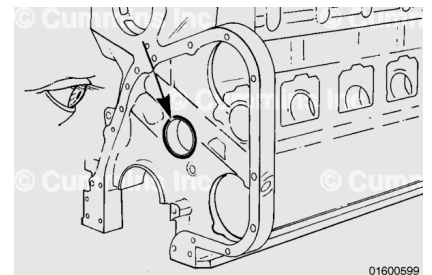
After the cutting operation is complete, use a rigid or flexible hone to remove any burrs from each camshaft bore.

Remove all burrs from the oil supply hole drilling. The surface finish of the camshaft bore after honing **must** be $2.03\text{ }\mu\text{m}$ [80 μin] or less.



Inspect the enlarged camshaft bore for any fault indication that was **not** discovered during the initial inspection before proceeding.

This is especially critical in instances where the bore was distressed by heat or impact loading.

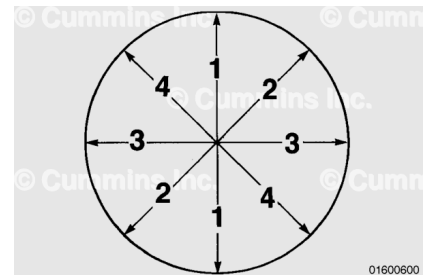


Note : If the repair exceeds these dimensions, the cylinder block can not be used as there are no larger repair sleeves available.

Check each camshaft bore to be sure that it is the correct size for the required press fit on the camshaft bushing repair assembly, or repair sleeve.

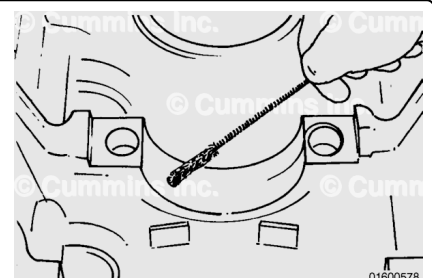
Measure each bore in four equally spaced directions.

Reference the beginning of the camshaft bore machine section for press fit specifications.



The block **must** be thoroughly cleaned and free of all machining debris.

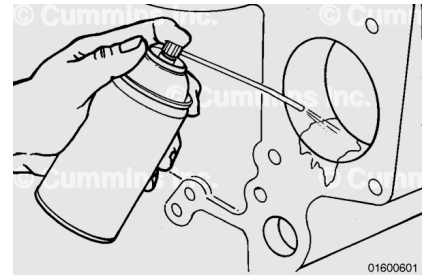
Be sure to brush the oil supply holes in the cylinder block with a nonmetallic brush to remove all machining debris.



Prepare the bore for installation of the repair sleeve.

Clean the bore with spray cleaner, Part Number 3375433 or ST-1272-13.

Allow 2 minutes for the excess cleaner to evaporate.



Clean the outside diameter of the repair sleeve.

Use spray cleaner, Part Number 3375433 or ST-1272-13, to remove any oils, greases, or preservative coating.

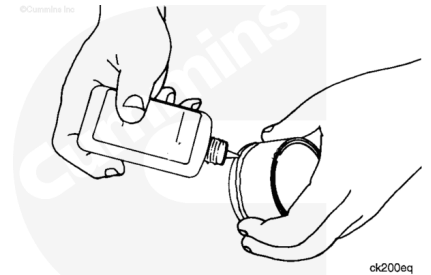


Use liquid nitrogen to freeze the repair sleeve.

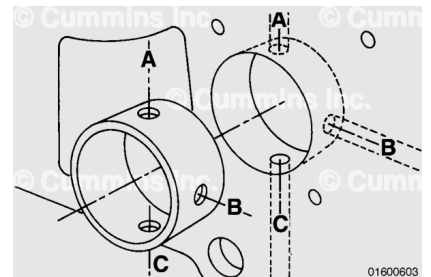
Apply a 0.8 mm [1/35 in] wide bead of Loctite™ 601 or equivalent to the leading edge of the repair sleeve.

Allow the bead to extend completely around the outside diameter of the repair assembly.

Each bead should be the full length of the repair assembly.

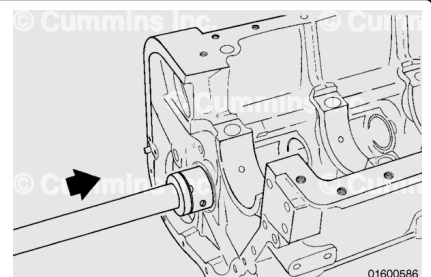


Turn the repair sleeve until the oil holes in the repair sleeve are correctly oriented with the oil supply drillings in the cylinder block camshaft repair bore.



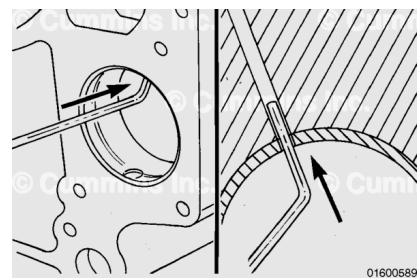
Press or drive the repair sleeve into the repair bore of the cylinder block.

Use installer camshaft bore insert kit, Part Number 3823621, to install the repair sleeve.

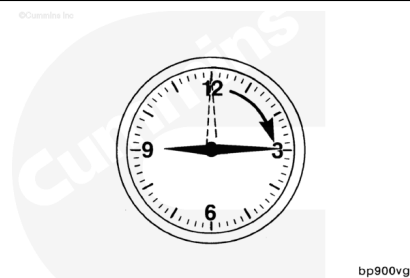


Check the location and position of the oil supply holes.

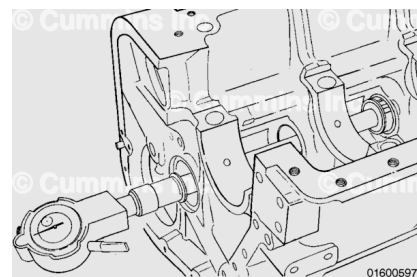
Be sure a 7-mm [1/4-in] Allen wrench will pass through the repair assembly oil holes and into the oil supply drillings in the cylinder block.



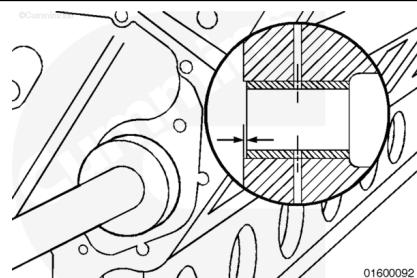
Allow the sealant to dry for 10 minutes.



Verify the camshaft bore in four equally spaced locations and compare with the camshaft bore specifications.



Install the camshaft bushing. Reference the Install section of this procedure.



If there is any resistance when installing the camshaft or when the camshaft is rotated, the camshaft bores are out of alignment. Engine damage will occur if a camshaft block with out-of-alignment bores is used.

Reference the following procedure to install the camshaft.
Refer to Procedure 001-008 in Section 1.

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